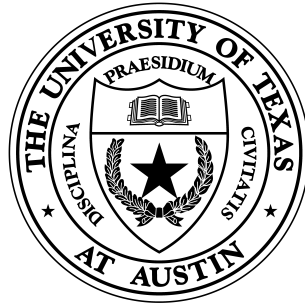


University of Texas at Austin

LIN 371: Machine Learning for Text Analysis
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NLP Mental Health Classification Model

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1 Background and Motivation

As college students, mental health is a huge issue that we believe could be better addressed. Every year, approximately 1100 college students commit suicide. According to the National Institute of Health, 12% of all college students have suicidal thoughts during their education (Wilcox, 2010). Addressing this starts with identifying those who are at risk of mental illness. This project is about understanding how certain words/phrases correspond to different mental health conditions. We will achieve this by creating a classifier that can identify when text contains depression, anxiety, or suicidal sentiment.

2 Data & Resources

The data originally comes from suicide related Reddit communities, and was compiled through Kaggle for mental health classification tasks.¹ The labels come from human and auto labeled sources, and may contain substantial noise. For instance, not everyone participating in communities regarding suicide are posting content with suicidal sentiment. Each data entry contains a unique ID, raw text, and one of the four labels: suicidal, anxiety, depression, or normal.

- **Training Data (39,689 entries)**

- Normal: 14,713 Depression: 11,605 Suicidal: 8,969 Anxiety: 4,402

- **Testing Data (9,923 entries)**

- Normal: 3,678 Depression: 2,901 Suicidal: 2,243 Anxiety: 1,101

- **Sample Entry**

- Input: “Trouble sleeping, confused mind, restless heart. All out of tune”
 - Output: Anxiety

Note that the data set is protected under Attribution 4.0 International licensing, which means that we are able to share and adapt as long as proper credit is given.

We utilized the Sci-kit Learn framework for text pre-processing and data training. The Count Vectorizer handled stop words and tokenization, and a stratified split between the train and test was used to handle unbalanced classes.

In terms of ethical considerations, the data does not contain Reddit usernames. In addition, the usage of this model should only be for educational and experimental purposes, not for diagnosis or labeling illness.

¹Subreddits include: r/suicidewatch, r/depression, and 13 others that are not explicitly listed.

	Logistic Regression	Naive Bayes	XGBoost
Recall	73.34%	68.93%	75.72%
Precision	74.10%	70.61%	76.49%
Accuracy	75.75%	69.13%	77.82%

Table 1: *These metric were calculated with "macro weighting" meaning that they were calculated for each class and then averaged together.*

It can be clearly seen that all three models vastly outperformed the majority class classifier. Overall, XGBoost performed the best of the three models in recall, precision, and accuracy. ³

4.1 Confusion Matrix

A confusion matrix will allow us to easily spot where our model made mistakes. This is done by looking at the value corresponding with the predicted label along the x-axis with the true label from the y-axis.

From the confusion matrix for XGBoost it is evident that the model often misinterpreted suicidal and depression sentiment. A positive takeaway from the matrix is that only 488 entries were falsely labeled as normal which is less than 5%. Another positive takeaway is that only 2.5% of suicidal labels were falsely labeled as normal, which would be the most detrimental outcome for our model.

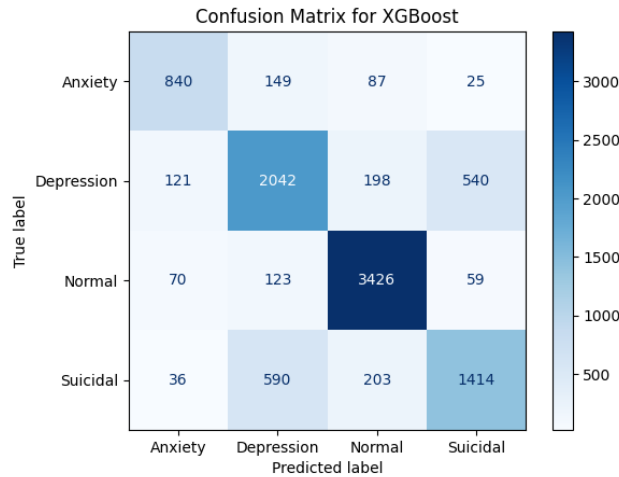


Figure 2: *Confusion matrix for logistic regression model.*

The confusion matrices for logistic regression and naive bayes can be found in the appendix at Figure 5 and 6.

4.2 Receiver Operating Characteristic

Finally, a Receiver Operating Characteristic (ROC) is used to evaluate performance for each class by plotting the True Positive Rate against the False Positive Rate. Since ROC is inherently binary, we used a One-vs-Rest (OvR) approach where the ROC curve for each class is plotted against all others. The Area Under the Curve (AUC) was then used to evaluate the model's ability to separate between classes, with values closer to 1 indicating better performance.

From the curve below, we can see that the XGBoost model does well in differentiating between each of the 4 classes. It does exceptionally well in understanding the normal and the anxiety classes,

³Figure 9 includes indepth information on XGBoost's performance by class.

and can slightly improve with the suicidal and the depression classes. Overall, these results show that instead of randomly guessing, the model understands the difference between each of the 4 classes.

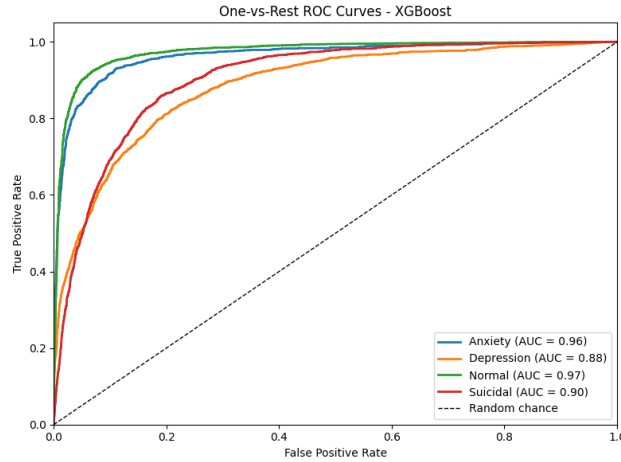


Figure 3: *Receiver operating characteristic curve for logistic regression model.*

The receiver operating characteristic curve for logistic regression and naive bayes can be found in the appendix at Figure 7 and 8.

5 Limitations

One limitation is that a text entry could represent multiple emotions. For instance, text that has been labeled for anxiety may also exhibit the traits of depression. As a result, the model can have some trouble differentiating between the Suicidal class and the Depression class.

Another limitation is that the data was compiled from multiple different datasets with different standards for labeling. The data consists of a mix of human labeled entries as well as data labeled based of the corresponding subreddit the post was found. This raises concerns regarding the credibility of the labels.

Below are some entries that the XGBoost model labeled incorrectly.

- The sad truth is. If I don't text people first I'll never hear from them again. It happens every single time... I can feel myself dying from the loneliness
– **Predicted:** Depression, **Actual:** Anxiety
- So, basic information, I have tried to suicide 3 times... any suggestion is good, I am open minded. Long story short
– **Predicted:** Depression, **Actual:** Suicide

6 Conclusion

This project preprocessed raw text, created visualizations to find patterns, and created a text classification pipeline to understand how certain phrases affect mental health conditions. Through the visualizations we saw that wants and feelings play an important role. In addition, we found that relationships are an important context. In the modeling portion we tested different models, discovering the XGBoost to be the most accurate in terms of recall, precision and accuracy. In terms of evaluation, the model could improve on interpreting between suicidal and depression entries. It performed well in identifying false negatives and differentiating between each of the 4 classes.

7 Roles

Max handled all data pre-processing, including cleaning, tokenization, and vectorization. Ibrahim handled data exploration, which included identifying trends through tables/visualizations and setting up the GitHub repository (README, License, etc.). Together we completed the modeling and evaluation/final report.

8 Citations

Mukherjee, Priyangshu. “Mental Health Text Classification Dataset.” *Kaggle.com*, 2021,

www.kaggle.com/datasets/priyangshumukherjee/mental-health-text-classification-dataset/data. Accessed 13 Mar. 2026.

Wilcox, Holly C., et al. “Prevalence and Predictors of Persistent Suicide Ideation, Plans, and Attempts during College.” *Journal of Affective Disorders*, vol. 127, no. 1-3, Dec. 2010, pp. 287–294, <https://doi.org/10.1016/j.jad.2010.04.017>.

9 Appendix

GitHub: https://github.com/Ibrahim581/LIN_371_Project/tree/main

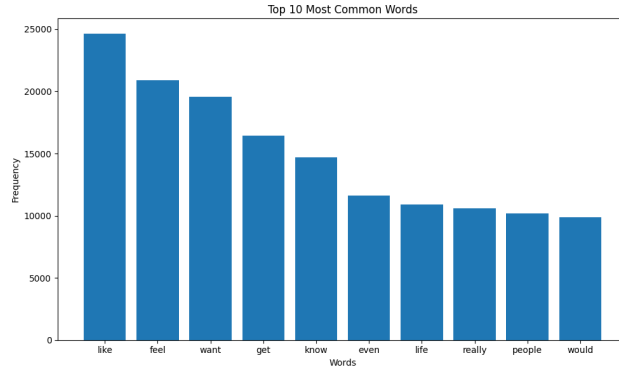


Figure 4: *Top ten most common words.*

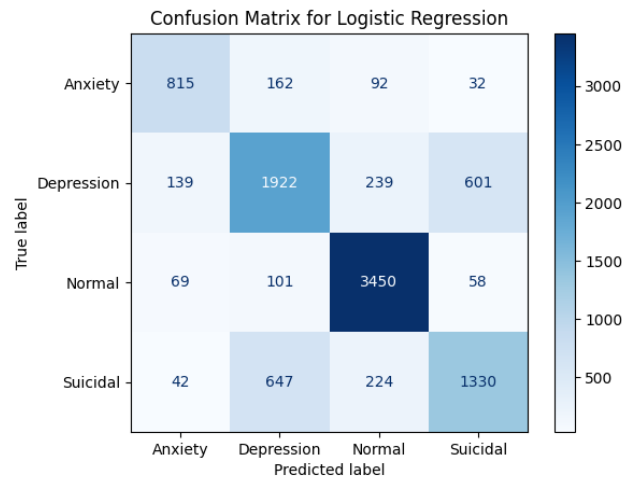


Figure 5: *Confusion matrix for logistic regression model.*

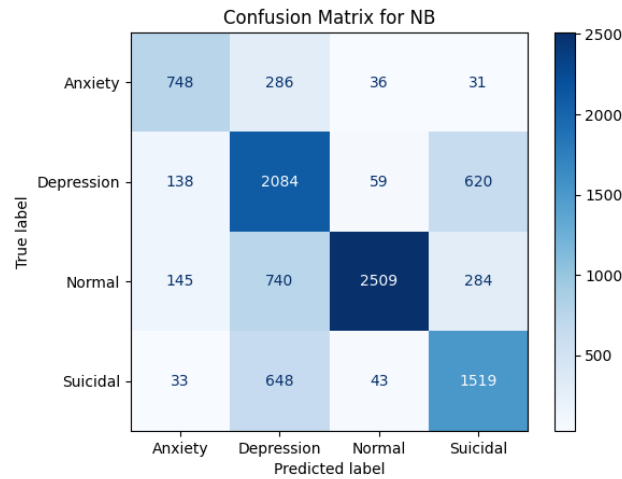


Figure 6: *Confusion matrix for naive bayes model.*

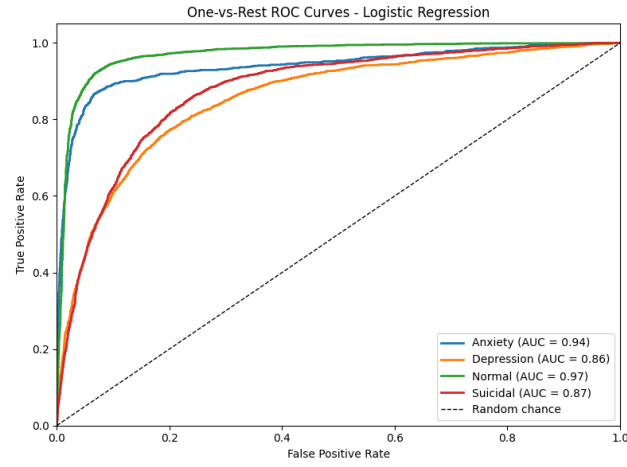


Figure 7: Receiver operating characteristic curve for logistic regression model.

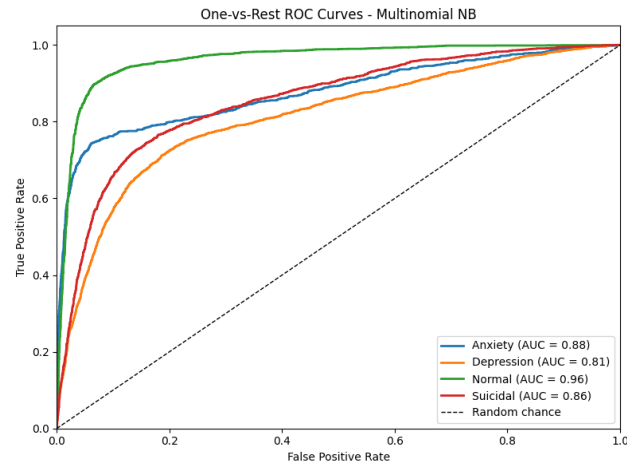


Figure 8: Receiver operating characteristic curve for naive bayes model.

	precision	recall	f1-score
Anxiety	0.79	0.76	0.77
Depression	0.70	0.70	0.70
Normal	0.88	0.93	0.90
Suicidal	0.69	0.63	0.66

Figure 9: XGBoost precision and recall by class.